Technical Appendix

Opening the Black Box of Large Language Models in Text-based Measurement: Staged Pipelines with Verifiable Checkpoints

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Proposal for Special Issue: AI-Enabled Frontiers in Organizational Science (*Organization Science*)

Repository: https://github.com/HC-Geopolitics-Innovation/LLM-Text-Analysis-and-Synthesis

Appendix A. Overview of Steps in Large Language Model Enabled Research Pipeline

A.1. Long-Document Text Summarization

A.1.1. Overview of Processing Pipeline

Processing long text documents is a challenge for large language models because they are prone to degradation in attention as input text increases (Liu et al., 2023), struggle to preserve factual detail across multiple conversational turns, and can introduce hallucinations during complex open-ended synthesis.

To overcome these constraints, we propose a staged Map-Reduce summarization pipeline that preserves document integrity:

1. **Data Ingestion & Speaker Turn Partitioning:** Documents are ingested from source XML/HTML formats, parsed into turn-level records, and attributed to specific speakers.

2. **Chunk-Level Map Summarization:** Documents exceeding the model's high-attention context threshold are divided into sequential semantic chunks (roughly 4,000–8,000 tokens) with overlapping boundaries to prevent conversational clipping. Prompts focus the model on identifying substantive factual points.

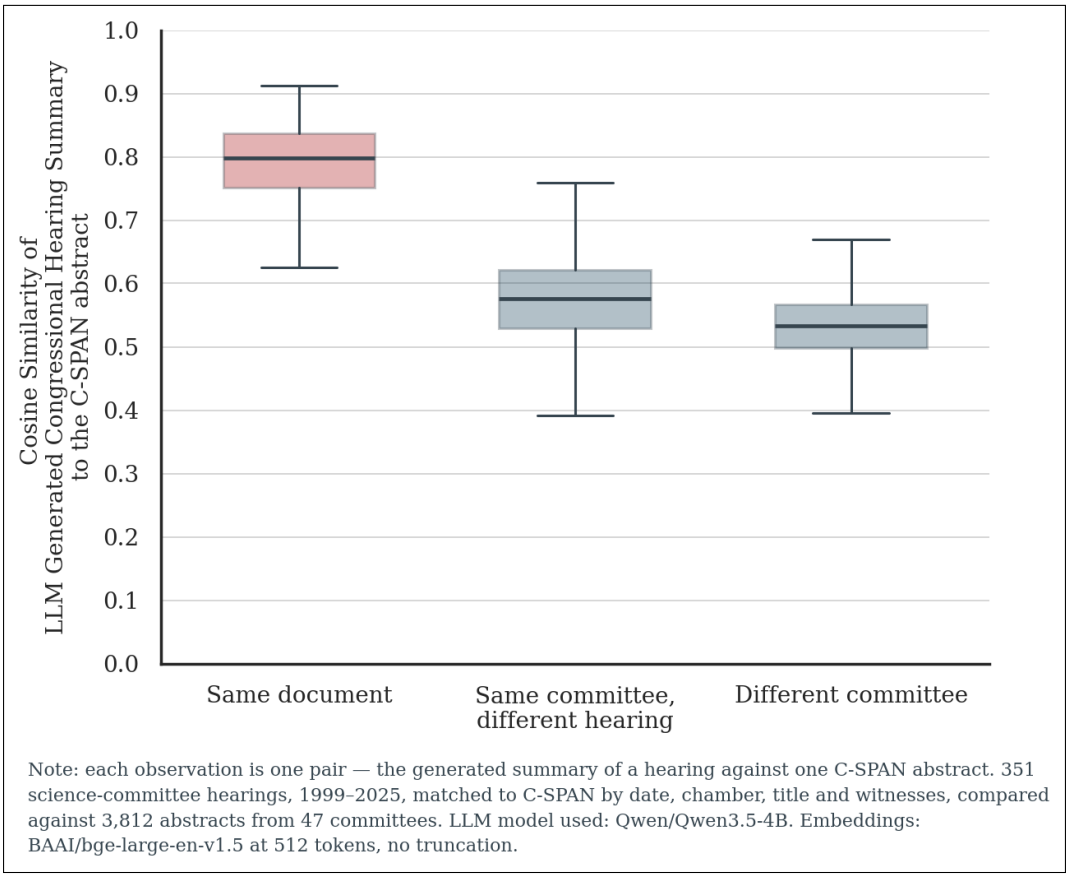
3. **Hierarchical Reduce Synthesis:** Intermediate chunk summaries are aggregated through a structured reduce prompt that reconciles cross-chunk themes into an entity-attributed document summary.

4. **Discriminant Validity Checkpoint:** Summaries are validated against external human benchmarks (such as professional C-SPAN editorial abstracts).

A.1.2. Application: Summarizing Long Text Congressional Hearings

We apply this architecture to 2,352 congressional committee hearings before the House Science, Space, and Technology Committee and the Senate Commerce, Science, and Transportation Committee. Transcripts average roughly 40,000 words. Figure 1 reports the cosine similarity between our generated summaries and independent, professional C-SPAN abstracts across same-hearing, same-committee, and different-committee pairs.

Figure 1: Cosine Similarity Between Hand-Coded Summaries and Large Language Model Generated Summaries across Committees



Note: each observation is one pair — the generated summary of a hearing against one C-SPAN abstract. 351 science-committee hearings, 1999–2025, matched to C-SPAN by date, chamber, title and witnesses, compared against 3,812 abstracts from 47 committees. LLM model used: Qwen/Qwen3.5-4B. Embeddings: BAAI/bge-large-en-v1.5 at 512 tokens, no truncation.

Notes: Comparison of cosine similarity distributions between model-generated summaries and independent human-authored C-SPAN abstracts. Same-document pairs exhibit high semantic alignment, whereas different-committee pairs show distinct separation, verifying discriminant validity.

A.1.3. Research Outputs

- **Cleaned Transcript Segments:** Fully segmented, speaker-attributed transcripts with timestamp and witness provenance.
- **Intermediate Chunk Summaries:** Auditable, passage-level summaries supporting localized verification.
- **Final Hearing Summaries:** Entity-attributed committee hearing summaries for downstream classification.

A.2. Extracting Topics, Positions, and Arguments from Long-Form Documents

A.2.1. Overview of Processing Pipeline

1. **Taxonomic Clustering & Codebook Generation:** We use LLMs to extract emerging policy topics from statements and crosswalk them to standardized classification systems like the Comparative Agendas Project (CAP).
2. **Stance Extraction:** Identifying discrete positions (support, concern, neutral) held by individual speakers on focal topics.
3. **Validation Against External Criteria:** Agreement audits across prompt variants and cross-validation against legislative roll-call voting records.

A.2.2. Application: Extracting Stances and Arguments around Focal Topics

Figure 2 illustrates the distribution of identified congressional stances across major technological domains, highlighting the balance of support versus concern.

A.2.3. Research Outputs

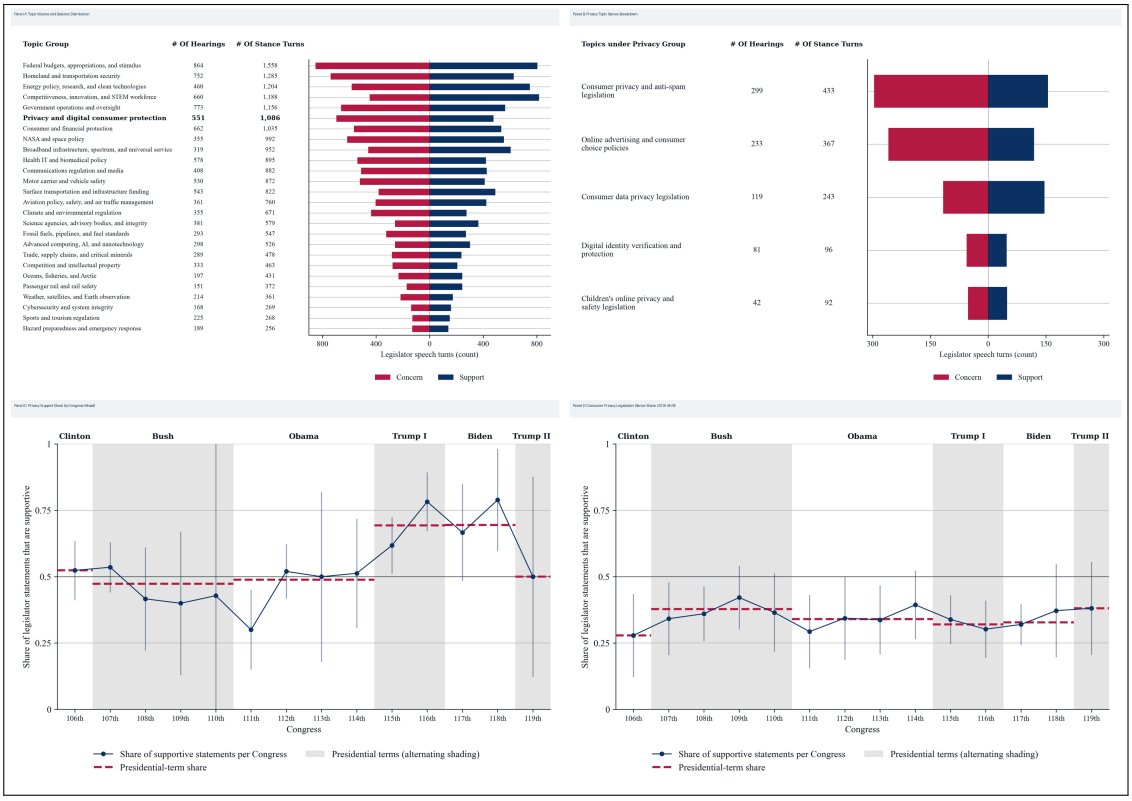
- **Standardized 111-Topic Policy Codebook:** Fully crosswalked to CAP categories.
- **Member-Topic Stance Panel:** Micro-level dataset tracking individual member arguments and policy positions over time.

A.3. Policy Relevance for Publicly Listed Firms

A.3.1. Overview of Processing Pipeline

1. **Corporate Text Extraction:** We extract Item 1 (Business Description) from annual 10-K filings.
2. **Centroid Embedding Estimation:** Generating high-dimensional dense embeddings of policy debate stances and firm business summaries.
3. **Cosine Distance Scoring:** Measuring semantic cosine similarity between each firm's business profile and the policy centroid, demeaned relative to sample benchmarks.

Figure 2: Topics and Overview of Stance (Support / Concern of Policymakers)



Notes:

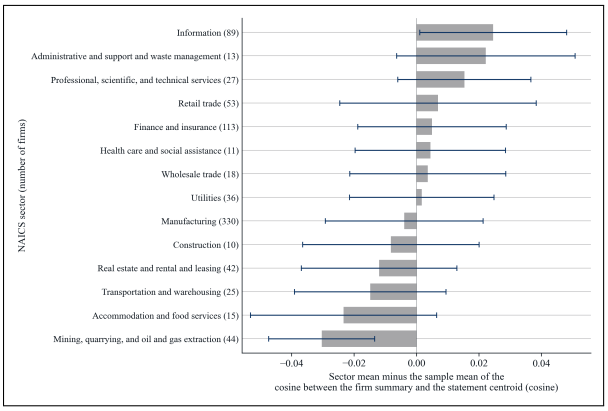
Four-panel overview tracking congressional topic prevalence, stance distributions, member polarization, and longitudinal legislative attention across science and technology hearings.

A.3.2. Application: Connecting Focal Concerns and Opportunities of Policymakers to Business Areas of Publicly Listed Firms

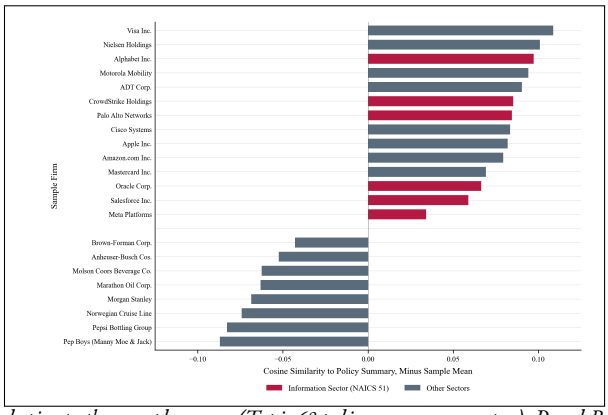
Figure 3 reports topic relevance scores for publicly listed firms regarding Consumer Data Privacy Legislation (Topic 68). Panel A reports industry-level exposure by NAICS sector relative to the sample mean. Panel B reports firm-level relevance scores for selected high- and low-relevance firms.

Figure 3: Topic Relevance Scores for Publicly Listed Firms

Panel A: NAICS Sector Cosine Relative to Mean



Panel B: Selected High and Low Relevance Firms



*Notes: Panel A reports firm summary cosine similarity by NAICS sector relative to the sample mean (Topic 68 policy summary vector). Panel B reports demeaned topic relevance scores for selected high- and low-relevance firms, color-coded by sector (Information sector in crimson, other sectors in slate grey). Both panels share identical aspect ratios (*1.5*).*

A.3.3. Research Outputs

- **10-K Item 1 Business Description Corpus:** Curated and verified annual report summaries.
- **Firm-Topic Relevance Panel:** Longitudinal exposure scores linking public firms to policy debates.

Appendix B. Overview of Datasets Used for Data Pipeline

This appendix provides an integrated overview of the primary input corpora, verification benchmarks, and research output datasets utilized across our AI-enabled pipeline.

B.1. Primary Data Inputs

- **Congressional Hearings Dataset (U.S. GPO GovInfo CHRG Collection):** Complete package manifest of 36,378 hearings spanning 1995–2025. Focal sample covers 2,352 hearings before House Science and Senate Commerce committees containing 93.6 million words across 116,000 speaker turns.
- **Publicly Listed Firms Business Descriptions (SEC Form 10-K Filings):** Full SEC EDGAR archive accessed via sec-api.io, comprising 305,140 annual reports. Mapped to financial fundamentals through a bijective CRSP–Compustat crosswalk linking 25,207 CIK $\leftrightarrow$ GVKEY identifier pairs. Yields 22,299 S&P 500 reports complete with extracted text and usability flags.

B.2. Verification Data

- **Independent Reference Abstracts (C-SPAN Archives API):** 18,697 committee program summaries covering 1995–2025 written by professional C-SPAN editors independently of congressional staff, serving as an external ground-truth benchmark for discriminant validity testing.

B.3. Output Research Datasets

- **Text Summarization Corpus (Stage 2 Outputs):** Cleaned, turn-attributed transcripts; intermediate chunk summaries; finalized hearing summaries (`hearing_summaries.parquet`).
- **Policy Stance Panel Dataset (Stage 3 Outputs):** Standardized 111-topic policy taxonomy codebook (`frozen_codebook_11`) longitudinal member-topic stance panel (`member_stance_panel.parquet`).
- **Corporate Topic Relevance Dataset (Stage 4 Outputs):** Form 10-K Item 1 corporate business description embeddings; congressional debate centroid vectors; firm-year policy exposure panel (`firm_topic_relevance_panel.parq`

Appendix C. Example of Research Enabled by LLM Data Pipeline

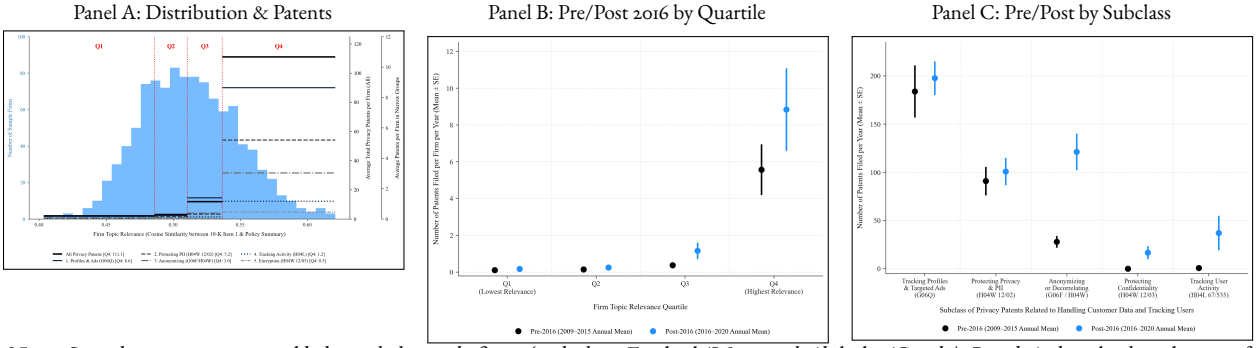
Evaluating the Impact of Privacy Policy Debate Following the Cambridge Analytica Scandal on the Trajectory of Firm Corporate Research on Privacy Technology

As described above, we demonstrate how the application of our data pipeline to congressional testimony and company 10-K filings allows us to quantify which firms were impacted by policy debate and study this question in a way not previously possible.

Leveraging this pipeline, we generate an index providing both the stance of a policy debate (corporate and consumer data privacy) and the extent to which these policy positions impacted specific firms. Crucially, these issues would not have been possible to identify, let alone empirically study, in the absence of our staged approach: while prominent technology firms like Meta/Facebook were known targets following the Cambridge Analytica scandal, identifying the broader universe of affected firms across diverse sectors has historically been intractable.

In Figure 4, we evaluate patenting intensity across topic relevance quartiles (Q_1 to Q_4). Panel A plots the distribution of topic relevance scores against the intensity of privacy patenting. Firms identified as more exposed to the privacy debate generate an overwhelming share of privacy patents. Panel B evaluates annual patent filing rates before and after 2016 (2009–2015 vs. 2016–2020), showing that privacy innovation surged predominantly among Quartile 4 firms (rising from 5.57 to 8.84 patents per firm per year, $+3.27$ patents/year/firm), while lower quartiles remained flat. Panel C disaggregates these changes across specific technology subclasses, revealing surges in data anonymization (`G06F 21/6254`, `H04W 12/033`, $+335.1\%$) and network-level activity tracking (`H04L 67/535`, $+5,080\%$).

Figure 4: Firm Topic Relevance and Privacy Patenting Responses Pre- and Post-2016



Notes: Sample comprises 1,058 publicly traded sample firms (excluding Facebook/Meta and Alphabet/Google). Panel A plots the distribution of firm topic relevance (cosine similarity between firm Form 10-K Item 1 business descriptions and the Congressional policy summary for Topic 68: Consumer Data Privacy Legislation; left axis, scaled 0–100). Red dotted vertical lines denote quartile boundaries ($Q_1 \leq 0.486$, Q_2: 0.486–0.511, Q_3: 0.511–0.537, $Q_4 > 0.537$). Right Axis 1 traces average total privacy patents per firm across 1999–2021 (rising from 2.2 in Q_1 to 111.1 in Q_4). Right Axis 2 traces average patents per firm across the five narrow technology subclasses: Tracking Profiles & Targeted Ads (G06Q 30/0269, 30/0255, 30/0261), Protecting Privacy & PII (H04W 12/02), Anonymizing Data (G06F 21/6254, H04W 12/033), Tracking User Activity (H04L 67/535), and Protecting Confidentiality Encryption (H04W 12/03). Panel B reports annual privacy patents filed per firm before (2009–2015 annual mean) and after (2016–2020 annual mean) by topic relevance quartile, with capless standard error bars. Panel C reports annual patent filings across the five technology subclasses before and after 2016.

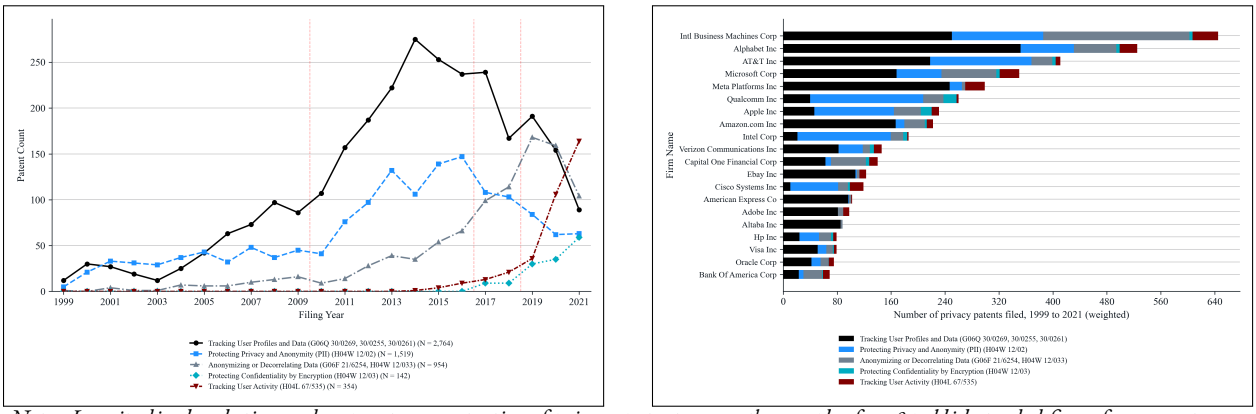
In Figure 5, we decompose these results further by investigating how these patterns appear in the raw patent data. In Panel A, we plot the number of patents filed in each class in each year and observe a large increase in patents related to tracking user profiles and collecting customer data, particularly increasing after 2009, coinciding with the widespread adoption of smartphones. This is mirrored by an increase (albeit less pronounced) of technology

designed to protect customer data. Not reflected in this figure, but in this period, the transmission of digital data through digital mobile networks saw an increase in patenting of patents related to encryption and data protection over mobile networks by hardware and infrastructure manufacturers. Panel B displays the 20 sample firms with the highest total privacy patent holdings over 1999–2021, broken down by technology class.

Figure 5: Long-Term Trends and Firm Concentration in Privacy Patenting (1999–2021)

Panel A: Annual Privacy Patents by Topic Group

Panel B: Top Privacy Patenting Firms

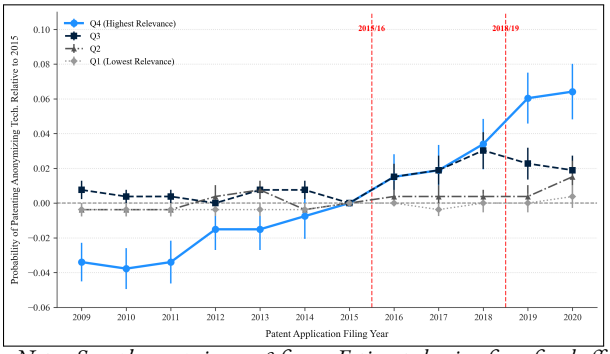


Notes: Longitudinal evolution and corporate concentration of privacy patents across the sample of 1,058 publicly traded firms from 1999 to 2021 matched to USPTO utility patent grants using CPC codes. Panel A plots annual patent counts across five focal technology classes: (1) Tracking User Profiles and Data (G06Q 30/0269, 30/0255, 30/0261); (2) Protecting Privacy and Anonymity in PII (H04W 12/02); (3) Anonymizing or Decorrelating Data (G06F 21/6254, H04W 12/033); (4) Protecting Confidentiality by Encryption (H04W 12/03); and (5) Tracking User Activity (H04L 67/535). Vertical dashed lines indicate regulatory milestones (2009.5, 2016.5, 2018.5). Panel B displays the 20 sample firms with the highest total privacy patent holdings over 1999–2021, with stacked bars indicating portfolio composition.

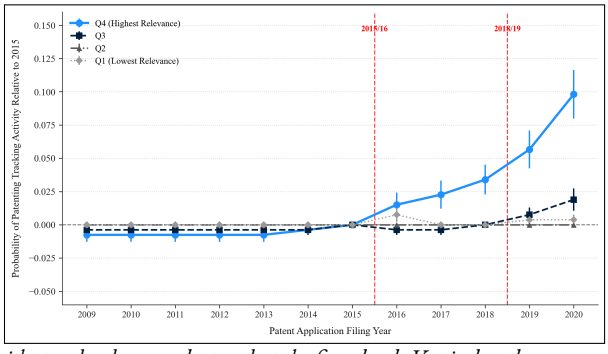
In Figure 6, we investigate this further by exploring how the probability of firms pursuing these technologies changed in relation to 2015. This is by no means an exogenous event: the increase in policy discussion coincided with a broader debate and pressure on how companies manage customer data. However, it provides a useful setting to evaluate the diagnostic usefulness of the relevance index. We present annual event-study estimates of patenting probability from 2009 to 2020 relative to 2015 across topic relevance quartiles (Q_1 to Q_4). The outcome in each panel is an indicator equal to one if firm i filed at least one patent application in year t in the focal technology group, estimated using firm fixed effects with standard errors clustered at the firm level.

Figure 6: Annual Event-Study Estimates of Patenting Probability Relative to 2015 by Relevance Quartile

Panel A: Anonymizing / Decorrelating Data



Panel B: Tracking User Activity



Notes: Sample comprises 1,058 firms. Estimated using firm fixed effects with standard errors clustered at the firm level. Vertical capless error bars represent 95% confidence intervals. Vertical red dashed lines indicate the 2015/2016 policy transition and the 2018/2019 GDPR/CCPA enforcement milestone. Panel A evaluates Anonymizing or Decorrelating Data (CPC G06F 21/6254, H04W 12/033); Quartile 4 firms exhibit a sharp increase of $+6.0$ *pp in 2019 (*$p < 0.001$*) and* $+6.4$ *pp in 2020 (*$p < 0.001$*). Panel B evaluates Tracking User Activity (CPC H04L 67/535); Quartile 4 firms experience a continuous rise up to* $+9.8$ *pp in 2020 (*$p < 0.001$*). Pre-trends are flat across all quartiles.*

Appendix D. Reproducible and Adaptable Data Pipelines with Coding Agents

While this data pipeline leverages AI tools to perform critical measurement tasks, an additional structural benefit is emerging: research pipelines shared via online repositories (like GitHub) in machine-readable formats can be rapidly deployed, verified, and adapted by autonomous coding agents.

Rather than treating papers as static documents, online repositories serve as active software environments where researchers share modular tools. To demonstrate this capability, the accompanying repository folder (`X2 - Example of Reproducing and Adapting the Research Pipeline/`) provides:

1. **Sample Agent Prompt:** An actionable prompt (`X1 - Sample Agent Prompt For Adapting Research Pipeline To New Question.md`) instructing an LLM coding agent to read the repository architecture, inspect input corpora, and adapt the staged methodology to a new research setting (e.g., regulatory debate surrounding renewable energy and corporate carbon emission innovation).
2. **Implementation Plan:** A full, researcher-evaluable implementation plan (`X2 - Sample Implementation Plan Generated by Pipeline.md`) detailing step-by-step adaptations, schema crosswalks (from CPC privacy codes to Y02 climate mitigation patent classifications), verification checkpoints, and execution timelines.

Researchers can provide this prompt to their own AI tools, evaluate the proposed implementation plan, and deploy adaptable pipelines for organizational and social science research.